上海交通大学学位论文

面向 3C 工业生产线的双臂机器人视觉-语言-动作模型

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A VISION-LANGUAGE-ACTION MODEL FOR DUAL-ARM ROBOTS IN 3C INDUSTRIAL PRODUCTION LINES

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摘 要

现代 3C 电子装配线需要处理小型元器件、高产品多样性和富接触操作，这些任务仍难以用传统固定程序机器人实现自动化。本项目针对这一缺口，面向双臂工业操作开发视觉-语言-动作（Vision-Language-Action, VLA）系统。该系统拟运行于双臂 Franka 平台，并覆盖典型 PC 关键部件任务，具体包括内存条插入、CPU 放置以及电源线连接。

当前实现基于 LeRobot 框架中的 Action Chunking with Transformers（ACT）策略，通过遥操作采集分阶段、任务相关的演示数据，并按目标任务部署对应检查点。现阶段验证聚焦于视觉-动作模仿学习基线；语言条件推理与统一多任务控制将作为后续扩展。最终概念验证目标为：规划成功率高于 90%，任务执行成功率高于 70%，部件检测精度高于 90%，位姿估计误差低于 20 mm，故障检测率高于 90%，恢复率高于 70%。

关键词： 双臂机器人，视觉-语言-动作模型，工业自动化

ABSTRACT

Modern 3C electronics assembly lines deal with small components, high product variety, and contact-rich operations that remain difficult to automate with conventional fixed-program robots. Our project targets this gap by developing a Vision-Language-Action (VLA) system for bimanual industrial manipulation. The system is designed to run on a dual-arm Franka platform and to handle representative PC key-component tasks—specifically RAM insertion, CPU placement, and power-cable connection.

The current implementation uses the Action Chunking with Transformers (ACT) policy in the LeRobot framework. It collects staged, task-specific demonstrations through teleoperation and deploys the checkpoint selected for the requested task. The present validation focuses on a vision–action imitation-learning baseline; language-conditioned inference and unified multi-task control remain later extensions. The final proof-of-concept targets are a planning success rate above 90%, a task execution success rate above 70%, component detection accuracy above 90%, pose estimation error below 20 mm, failure detection rate above 90%, and recovery rate above 70%.

Key words: dual-arm robot, vision-language-action model, industrial automation

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Chapter 1 Introduction

1.1 Background and significance

This chapter introduces the project background, motivation, and overall scope of developing a vision-language-action model for dual-arm robots in 3C industrial production lines.

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Bill of materials Appendix 2

Table 2–1 Bill of materials

Quantity	Part Description	Purchased From	Part Number	Price (each)	Notes
...	...	...	...	...	...

Research Projects and Publications during Undergraduate Period

[1] ……

Acknowledgements

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A VISION-LANGUAGE-ACTION MODEL FOR DUAL-ARM ROBOTS IN 3C INDUSTRIAL PRODUCTION LINES

HCCI (Homogenous Charge Compression Ignition) combustion has advantages in terms of efficiency and reduced emission. HCCI combustion can not only ensure both the high economic and dynamic quality of the engine, but also efficiently reduce the NOx and smoke emission. Moreover, one of the remarkable characteristics of HCCI combustion is that the ignition and combustion process are controlled by the chemical kinetics, so the HCCI ignition time can vary significantly with the changes of engine configuration parameters and operating conditions. ⋯⋯